

Subiect nr 16

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Grupa: 232

16.1. a) Deduceti iteratie care rezulta aplicand metoda lui Newton functiei

$$f(x) = x^3 - a = 0 \text{ pentru a calcula radacina cubica } \alpha = a^{\frac{1}{3}} \text{ lui } a > 0.$$

Solutie:

$$f(x) = x^3 - a$$

$$f'(x) = 3x^2 > 0$$

$$f''(x) = 6x > 0 \text{ pt } x > 0$$

$$x > 0$$

$$\text{In cazul general iteratia are forma: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\text{Inlocuind cu valorile concrete de mai sus obtinem: } x_{n+1} = x_n - \frac{x_n^3 - a}{3x_n^2}$$

$$\text{Deci } x_{n+1} = \frac{1}{3} \left(2x_n + \frac{a}{x_n^2} \right)$$

Pentru a alege un punct de pornire (x_0) observam ca $f(0) = -a < 0$

si $f(a+1) = (a+1)^3 - a > 0$ si $f''(x) > 0$ pt $x > 0$ deci o posibila valoare ar fi

$$x_0 = a+1$$

16.1. b) Considerati ecuatia echivalenta $f_\lambda = 0$ unde $f_\lambda(x) = x^{3-\lambda} - ax^{-\lambda}$, si

Determinati λ astfel incat metoda lui Newton sa converga cubic. Scrieti iteratie obtinuta in cea mai simpla forma.

Solutie:

$$\text{Consideram functia } \varphi(x) = x - \frac{f_\lambda(x)}{f'_\lambda(x)} \text{ deci, } \varphi(x) = x - \frac{x^{3-\lambda} - ax^{-\lambda}}{(3-\lambda)x^{2-\lambda} + \lambda ax^{-1-\lambda}}$$

Stim ca daca $\varphi'(\alpha) = \varphi''(\alpha) = \dots = \varphi^{(p-1)}(\alpha) = 0$ si $\varphi^{(p)}(\alpha) \neq 0$ atunci gradul de convergenta este p. (unde α - punct fix pt φ).

Cum $x^{-\lambda}(x^3 - a) = 0 \Rightarrow x = 0$, si $x = a^{\frac{1}{3}}$ sunt cele doua puncte fixe ale lui φ .

$$\varphi'(x) = 1 - \frac{4x^3 - a}{x^3(3-\lambda) + a\lambda} + \frac{3(x^4 - ax)x^2(3-\lambda)}{(x^3(3-\lambda) + a\lambda)^2}$$

$$\varphi''(x) = -\frac{12x^2}{x^3(3-\lambda) + a\lambda} + \frac{6(4x^3 - a)x^2(3-\lambda)}{(x^3(3-\lambda) + a\lambda)^2} - \frac{18(x^4 - ax)x^4(3-\lambda)^2}{(x^3(3-\lambda) + a\lambda)^3} + \frac{6(x^4 - ax)x(3-\lambda)}{(x^3(3-\lambda) + a\lambda)^2}$$

$$\varphi'''(x) = -\frac{24x}{x^3(3-\lambda) + a\lambda} + \frac{108x^4(3-\lambda)}{(x^3(3-\lambda) + a\lambda)^2} - \frac{54x^4(3-\lambda)^2(4x^3 - a)}{(x^3(3-\lambda) + a\lambda)^3} + \frac{18x(3-\lambda)(4x^3 - a)}{(x^3(3-\lambda) + a\lambda)^2} -$$

$$\frac{(x^4 - ax)162x^6(3-\lambda)}{(x^3(3-\lambda) + a\lambda)^4} - \frac{108(x^4 - ax)x^3(3-\lambda)^2}{(x^3(3-\lambda) + a\lambda)^3} + \frac{6(x^4 - ax)(3-\lambda)}{(x^3(3-\lambda) + a\lambda)^2}$$

Consideram cazul $\alpha = a^{\frac{1}{3}}$

Impunem conditia $\varphi'(\alpha) = 0 \Rightarrow 1 - \frac{3a}{a(3-\lambda) + a\lambda} = 0$ este adevarata pt orice λ .

Impunem conditia $\varphi''(\alpha) = 0 \Rightarrow -\frac{12a^{\frac{2}{3}}}{a(3-\lambda) + a\lambda} + \frac{18a^{\frac{5}{3}}(3-\lambda)}{(a(3-\lambda) + a\lambda)^2} = 0 \Rightarrow \lambda = 1$

Considerand $\lambda = 1$ obtinem $\varphi'''(\alpha) = \frac{4}{a^{\frac{2}{3}}} \neq 0$

Din calculele precedente deducem ca $p=3$, deci metoda lui Newton va converge cubic

pentru a calculat radacina $a^{\frac{1}{3}}$.

Pentru cazul $\alpha = 0$ constatam ca $\varphi'(\alpha) = 0 \Rightarrow \lambda = -1$ dar $\varphi''(\alpha) = 0$ si $\varphi'''(\alpha) = 0$ deci metoda lui Newton nu va converge cubic.

Pentru $\lambda = 1$, $f_1(x) = x^2 - \frac{a}{x}$ (cu $x \neq 0$), $f_1'(x) = 2x + \frac{a}{x^2}$, $f_1''(x) = 2 - \frac{2a}{x^3} > 0$ pt $x > a^{\frac{1}{3}}$

Se aplica interatia $x_{n+1} = \varphi(x_n)$ si se obtine:

$$x_{n+1} = x_n - \frac{x_n^2 - \frac{a}{x_n}}{2x_n + \frac{a}{x_n^2}} \text{ deci } x_{n+1} = \frac{x_n^2 + \frac{2a}{x_n}}{2x_n + \frac{a}{x_n^2}}$$

$f_1(a+1) > 0$ deci $x_0 = a+1$ ar putea fi punctul de plecare.